Master's Research Project:

Novel strategy to reposition drugs with a bioinformatics approach for treating Parkinson's

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Abstract

Introduction: Parkinson's disease (PD) is a neurodegenerative disease characterised by the progressive loss of dopaminergic (DA) neurons in the substantia nigra pars compacta (SNpc), resulting in impaired motility. Without comprehensive understanding of its pathogenesis, existing medications have been symptoms-relieving, presenting a need for developing disease modifying drugs. Multiple factors contributing to the susceptibility of DA neurons in SNpc appear to converge on the mitochondrial function. PINK1/Parkin is a key mitochondrial pathway in ensuring mitochondrial quality control. Alongside the growing enthusiasm in drug repurposing, this project aims to use a targeted repositioning strategy with a bioinformatics approach to uncover potential PD drugs that are disease modifying by targeting the PINK1/Parkin mitochondrial pathway.

Methods: A four-step process was carried out to shortlist potential drugs. Firstly, two transcriptional databases (CMap and LINCS) were interrogated to obatin compounds that upregulate the corresponding genes, PINK1 and PARK2. Secondly, these were shortlisted based on scores indicating the degree of upregulation. Thirdly, literature-based scrutiny was conducted to rule out compounds unfit for treating PD specifically. Finally, drugs without current clinical utility were eliminated, resulting in the final drugs: ethotoin, oxybutynin, and sirolimus. MCF-7 cells were exposed to the drugs for 24 hours before PINK1 and Parkin expression was determined by western blotting (n=6). Unpaired student's t-test was used for analysis.

Results: 33 compounds that upregulate PINK1/PARK2 were identified after the first two shortlisting steps, followed by a further drop of 24 that are unsuitable to be PD treatments, giving an intermediate list of 9 candidates. The final step eliminated 6 non-clinically used drugs, resulting in the 3 final drugs. Following drug exposure, Parkin expression was significantly higher compared to each vehicle control in MCF-7 cultures (ethotoin: $P<0.0001$; oxybutynin: $P=0.0003$; sirolimus: $P<0.0001$). Only sirolimus resulted in a significant increase in PINK1 expression levels ($P<0.0001$).

Conclusions: All three drugs showed the changes in PINK1/Parkin production in vitro that corresponded to the predicted upregulation of PINK1 and PARK2 from the databases. Although ethotoin was the most promising finalist, all three should be further analysed for their neuroprotective potential in animal PD models.

Abbreviations

α-syn, α-synuclein; BBB, blood-brain barrier; BCA, bicinchoninic acid; BD2K, Big Data to Knowledge; CMap, Connectivity Map; CNS, central nervous system; COVID-19, coronavirus disease 2019; DA, dopaminergic; DCIC, Data Coordination and Integration Centre; DJ-1, protein deglycase; DMEM, Dulbecco's Modified Eagle Medium; FBS, foetal bovine serum; GAPDH, glyceraldehyde 3-phosphate dehydrogenase; GLP, glucagon-like peptide; IMM, inner mitochondrial membrane; iPSCs, induced pluripotent stem cells; KO, knockout; LB, Lewy Body; LINCS, Library of Integrated Network-based Cellular Signatures database; LRRK2, leucine-rich repeat kinase 2; MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; MQC, mitochondrial quality control; 6-OHDA, 6-hydroxydopamine; OMM, outer mitochondrial membrane; PARIS, Parkin Interacting Substrate; PARK2, gene which code for Parkin; PBS, phosphate-buffered saline; (f/s)PD, (familial/sporadic) Parkinson's disease; PDD, PD-related dementia; PINK1, PTEN-induced putative kinase 1; P/S, penicillin-streptomycin; SNpc, substantia nigra pars compacta; SPIED3, Searchable Platform-Independent Expression Database III; USP, United States Pharmacopeia; WB, western blotting

1 Introduction

1.1 Background

1.1.1 Parkinson's Disease

Parkinson's disease (PD) is a neurodegenerative disorder characterised by the degeneration of dopaminergic (DA) neurons in the substantia nigra pars compacta (SNpc), resulting in a core triad of symptoms: resting tremor, bradykinesia, and elevated resting tone (Chakraborty et al., 2020; Ge et al., 2020). Apart from motor deficits, non-motor symptoms are not uncommon in PD, particularly in late-stage PD, such as sleep impairment and neuropsychiatric symptoms, including depression, anxiety, cognitive impairment progressing to dementia (Chakraborty et al., 2020).

The most widely known drug for treating PD is Levodopa, also known as L-DOPA, which acts as the precursor to dopamine neurotransmitter, to replenish reduced synaptic dopamine levels resulted from the loss of DA neurons (Chakraborty et al., 2020). Other pharmacological treatments have evolved to target mechanisms other than via replenishing dopamine, which include antioxidants, bioenergetic, trophic, anti-apoptotic and anti-inflammatory drugs, (AlDakheel et al., 2014; Chakraborty et al., 2020; Mor & Murphy, 2020). With the incomplete understanding of the pathogenesis of PD being a complex and multifactorial disorder (Rouaud et al., 2020), reluctantly, none of the existing treatments are disease modifying, and there is no cure for PD at present (Chakraborty et al., 2020). Nevertheless, the global clinical burden has more than doubled over the past generation as a result of increased life expectancy, better environmental conditions, such as exposition to the pollutant 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) and pesticides (Ferreira et al., 2020). There is, therefore, a growing unmet need for disease modifying treatments that intervene disease progression mechanisms; efforts have also been made in search of drugs with disease modifying potentials.

One approach that has received growing attention is drug repurposing as it is increasingly appreciated as a more efficient alternative to the costlier and slower classical drug discovery and development model. The enthusiasm in drug repurposing is valid, due to the higher probability of more rapid drug development with established compounds having identified molecular targets and primary indications in which they have been approved (Strittmatter, 2014). A repurposed drug is typically affordable, widely available, and often indicated for a common condition, such as diabetes (Tran & Prasad, 2020). As a matter of fact, exenatide, a glucagon-like peptide (GLP)-1 analogue, originally developed as a diabetic drug, was found to be suitable for treating PD with its neuroprotective and potential disease modifying effects shown in PD animal models via promoting mitochondrial biogenesis (AlDakheel et al., 2014; Mor & Murphy, 2020). Currently, it is undergoing phase III clinical trials as a potential disease modifying treatment for PD, such as Exenatide-PD3 (clinicaltrials.gov, NCT04232969).

Most recently, our lab has identified potential drugs that increase fibroblast growth factor 20 production, which exhibited neuroprotective effects in rats (Fletcher et al., 2019) in which the targeted repositioning strategy" was used, wherein a specific "target" involved in the disease is identified for drugs to be repurposed to be aiming at treating. An example of this "target" would be a key pathway contributing to the pathogenesis of a disease that shows therapeutic potential when being altered in a specific manner. In light of this, my project aims to use this targeted repositioning strategy to unravel potential PD therapeutics that are disease modifying. To achieve this, the first project aim is to identify a pathway involved in PD pathogenesis, in which a specific directional change in the expression of (a) target protein(s) is/are anticipated to show therapeutic potential for the treatment of PD. After conducting literature research, the mitochondrial PINK1/Parkin pathway was one that has been identified as critical in PD. With the potential of its intervention giving rise to a disease modifying treatment for PD, it was selected as the target for drug repositioning in this project.

1.1.2 Mitochondrial dysfunction in PD

There are multiple PD pathways that show a potential for drug targeting, including α-synuclein (α-syn) misfolding and aggregation, neuroinflammation and astrogliosis, mitochondrial oxidative stress, endo-lysosomal function, and more (Ge et al., 2020; Rouaud et al., 2020; Song et al., 2016). Nevertheless, current pharmacological interventions, rather than tackling PD pathophysiology and halting the disease progression, are predominately symptoms-relieving by nature and used for management, without a full understanding of its pathogenesis (Rouaud et al., 2020).

While the exact aetiology of PD has yet to be fully characterised, investigation into gene mutations in familial PD (fPD) have acted as pointers to proteins and systems that contribute to PD pathogenesis (Ryan et al., 2015). Converging evidence has suggested a clear contributing role of mitochondrial abnormalities and dysfunction in neurodegeneration and PD pathogenesis (Kazlauskaite & Muqit, 2015; Mor & Murphy, 2020; Zeng et al., 2020), in both fPD and sporadic PD (sPD) (Luo et al., 2015). PD-causing mutations have been shown to increase mitochondrial damage or impair clearance of damaged mitochondria due to stress (Ryan et al., 2015). Examples of common gene mutations in autosomal recessive fPD include PINK1, PARK2 and PARK7, which encode proteins with well-defined roles in maintaining mitochondria homeostasis and apoptosis (Ryan et al., 2015). Aside from genes in autosomal recessive fPD, mutations in other cytosolic proteins that cause autosomal dominant PD, such as leucine-rich repeat kinase 2 (LRRK2) and α-syn, have also been shown to modulate mitochondrial function (Ryan et al., 2015). Moreover, the propagation of α-syn prion-like behaviour proposed to be a key contributor to PD pathogenesis has been suggested to be associated with mitochondrial dysfunction and oxidative stress (Ryan et al., 2015).

Furthermore, in the specific context of neurons, their high bioenergetic demands imply a relatively higher degree of mitochondrial stress than average. This highlights the particular importance of mitochondrial quality control (MQC) in the resilience of cell populations and the maintenance of neuronal survival (Ge et al., 2020). Thus, MQC also otherwise infers the source of selective vulnerability in certain neuronal subpopulations, more vulnerable to developing neurodegeneration and Lewy Body (LB) pathology characteristic in PD, such as SNpc DA neurons (Ge et al., 2020; Ryan et al., 2015). In fact, mitochondrial size in SNpc DA neurons were found to be smaller than in neighboring non-DA neurons, or in DA neurons other brain regions more resilient to PD such as the ventral tegmental area (Luo et al., 2015).

While neurons are lost from a number of regions in PD patients, it is the vulnerability profile of SNpc neurons that resulted in the associated PD symptoms, characterising PD. Susceptibility factors include bioenergetic demands, increased oxidative stress, which converge on mitochondrial function (Ryan et al., 2015), collectively pointing to the role of mitochondrial dysfunction as a determinant of the susceptibility of SNpc DA neurons in PD.

1.1.3 PINK1/Parkin pathway

PINK1/Parkin pathway is a mitochondrial pathway acting as the first step in the activation of downstream MQC pathways in response to mitochondrial damage (Ge et al., 2020), such as by initiating mitophagy (Rouaud et al., 2020). As aforementioned, the importance of MQC in neurons is derived from their high bioenergetic demands, highlighting the importance of the presence and normal expression of functional PINK1 and Parkin proteins in maintaining neuronal survival under stress conditions (Lee et al., 2017; Luo et al., 2015; Mor & Murphy, 2020).

PINK1, PTEN-induced putative kinase 1, phosphorylates a range of substrates. Parkin, an E3 ubiquitin ligase in the cytosol, the protein encoded by the PARK2 gene, is one of the substrates that are phosphorylated and activated by PINK1 (Ge et al., 2020; Shin et al., 2011). Activated Parkin binds and catalyses the ubiquitination of downstream substrates (Ge et al., 2020; Kazlauskaite & Muqit, 2015; Lee et al., 2017; Luo et al., 2015), including protein deglycase (DJ-1), the protein encoded by PARK7 (Ryan et al., 2015). Another important downstream molecule with which Parkin interacts is Parkin Interacting Substrate (PARIS, otherwise known as ZNF746). PARIS represses the expression of PGC-1α, which demonstrates a neuroprotective role as the master regulator of mitochondrial biogenesis.

Under stress conditions, such as mitochondrial complex dysfunction and mutagenic stress, PINK1/Parkin acts as molecular switch transmitting the damage signal for subsequent events – mitophagy or mitochondrial biogenesis (Ryan et al., 2015). Normally, upon mitochondrial injury, PINK1 accumulates on the outer mitochondrial membrane (OMM) and activates Parkin to increase its ligase activity via two mechanisms: phosphorylating an ubiquitin that stabilises Parkin in its active conformation, and directly phosphorylating Parkin. Thus, Parkin is the amplifier of the damage detection signal from PINK1, the sensor of mitochondrial damage (Ge et al., 2020).

Involved in one of the key downstream pathways regulated by PINK1/Parkin activation, PARIS levels are kept low in healthy mitochondria with functional Parkin catalysing the degradation of PARIS, thereby allowing non-inhibited PGC-1α to stimulate mitochondrial biogenesis where more energy is needed under stress conditions. However, in pathological conditions, the loss of PINK1/Parkin results in the accumulation of PARIS, inhibiting PGC-1α expression and hence also mitochondrial biogenesis, where it is needed. Figure 1 shows a schematic of mitochondrial PINK1/Parkin pathway and their regulation of the downstream PARIS/PGC-1α pathway in healthy and pathological (PINK1/Parkin loss) stress conditions.

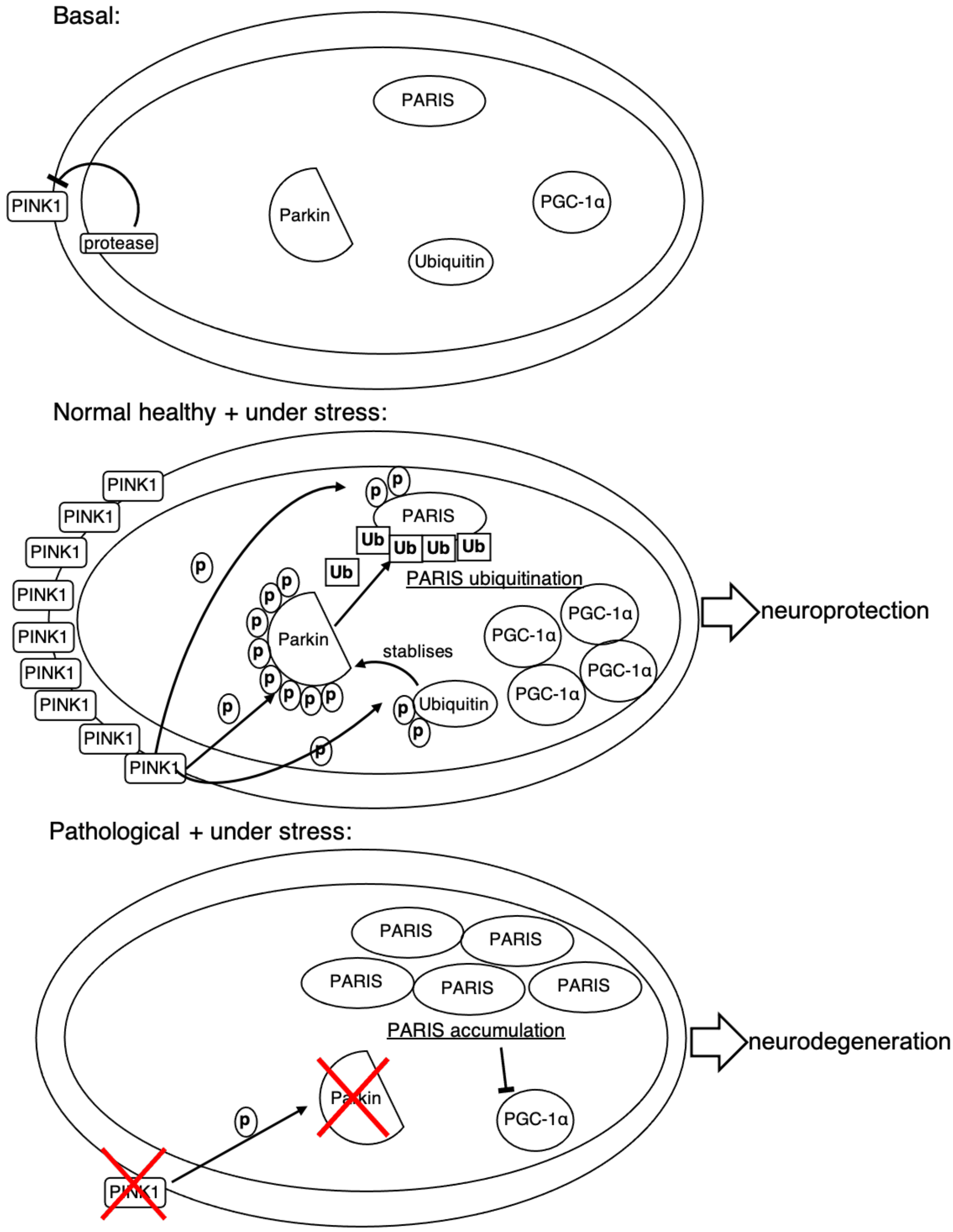

Figure 1. PINK1/Parkin mitochondrial pathway A schematic of the mitochondrial PINK1/Parkin signalling pathway and their regulation of the downstream PARIS/PGC-1α pathway in normal and pathological (PINK1/Parkin loss, indicated in red) stress conditions. In basal conditions, PINK1 expression is minimal due to cleavage by protease in the inner mitochondrial membrane. Normally under stress, PINK1 phosphorylates Parkin E3 ubiquitin ligase, which catalyses the ubiquitination of downstream molecules, including PARIS. PARIS accumulation inhibits of PGC-1α expression, which results in the PARIS-dependent neurodegeneration, as it disables PGC-1 from stimulating mitochondrial biogenesis where it is required under stress.

Of note, the PARIS/PGC-1α pathway appears to play a role in sporadic PD (Lee et al., 2017; Shin et al., 2011). Also, PARIS accumulation was found in human PD brain (Lee et al., 2017; Shin et al., 2011). PINK1/Parkin loss was also shown to cause circadian and sleep defects in hypothalamic neurons differentiated from PD patient induced pluripotent stem cells (iPSCs) and in fruit flies (Valadas et al., 2018).

In experiments, PARIS accumulation could be resulted from PINK1 depletion (Lee et al., 2017), as well as from the loss of Parkin in conditional knockout(KO)-PARK2 animal models (Shin et al., 2011), which both resulted in progressive PARIS-dependent DA neurons loss due to repressed PGC-1α expression. In phenotype rescuing experiments, Parkin overexpression has reversed the KO-PINK1 phenotype of PARIS-induced selective loss of DA neurons in SNpc (Kazlauskaite & Muqit, 2015; Shin et al., 2011). Mitochondrial fragmentation and dysfunction was also rescued by overexpression of PINK1 and Parkin (Luo et al., 2015).

Both positive and negative evidence has converged on the therapeutic potential of the overexpression of PINK1 and Parkin.

1.1.4 Further justification of choice of PINK1/Parkin pathway

Although the loss of PINK1/Parkin takes up a small fraction of fPD (Ge et al., 2020) and the majority (85%) of PD cases are sporadic (Ryan et al., 2015), the importance of MQC and thus the PINK1/Parkin pathway may represent a role of the pathway in the PD pathogenesis of a wider pool of sPD cases.

In addition, Parkin also has a role in regulating the endo-lysosomal pathway (Song et al., 2016), which is distinct from the mitochondrial PINK1/Parkin pathway, suggesting a potentially more widespread effect of altering Parkin expression in the global mitochondrial environment. This addresses a prior concern of the negligible effect of changes in the expression of few target protein(s).

Overall, the PINK1/Parkin could be an apt target for applying the targeted repositioning strategy in hopes of unravelling potential disease modifying drugs for the treatment of PD.

1.2 Project aims

My project aims at unravelling disease modifying treatments for PD by utilising a bioinformatics strategy to reposition potential drugs that upregulate PINK1/PARK2 to increase PINK1/Parkin expression. Derived from the rescue experiments of KO phenotypes by PINK1/Parkin overexpression, it is hypothesised that drugs that could increase in PINK1/Parkin production would be therapeutically useful in treating PD.

Having identified PINK1/Parkin pathway as the target pathway in the study, this project aims to use a bioinformatics approach to shortlist potential PINK1- and/or PARK2 boosting drugs according to the assigned criteria, such as the drug's suitability for chronic use in the treatment of PD and its current clinical utility.

Finally, the effects of the shortlisted drugs are tested in vitro to assess the translational potentials between the drug-induced transcriptional changes and the predicted increase in PINK1/Parkin protein expression.

2 Materials and Methods

2.1 Bioinformatics screening

In silico interrogation of transcriptional databases was carried out to screen for drugs that are predicted to increase PINK1 and PARK2 transcription. The bioinformatics tool, Searchable Platform-Independent Expression Database III (SPIED3) (Williams, 2013), was used as an integrated point of access to large-scale perturbation databases – the Connectivity Map 2.0 database (CMap) from the Broad Institute (Lamb et al., 2006), and the Library of Integrated Network-based Cellular Signatures database (LINCS) originally developed by the Big Data to Knowledge (BD2K) Data Coordination and Integration Centre (DCIC) (Keenan et al., 2018), later deposited by NCBI GEO (GEO accessions: GSE70138 and GSE92742). CMap consists of the gene expression profiles modified by 1309 drug-like compounds gathered from profiled on human cancer cell lines, including the MCF-7 human breast carcinoma cell line, whereas LINCS contains 15646 drug-like compounds profiled in 98 human cell lines.

Briefly, for CMap, profiles are based on array probes ranked according to expression relative to control with ranks going from 1 to ~25000, ~6000 of which corresponding to replicate treatments of the 1309 drugs. On SPIED3, ranks for these replicates were scaled to the range -1 to 1, combined for the same drug regardless of cell type, and filtered based on significance using a one sample student's t-test. The gene expression change ranks are defined as = $1 - 2\frac{R - R_{min}}{R_{max} - R_{min}}$, where R is the rank of the regulation of a given gene (R_{max} = the highest ranks, R_{min} = the lowest ranks) (Fletcher et al., 2019).

The data deposited by the LINCS consortium corresponds to each gene's level relative to plate average and SD in the form of Z scores ($Z = \frac{x - \mu}{\sigma}$, where x = rank of the regulation of a given gene, μ = mean, and σ = SD). Again, Z scores were combined for replicate treatments, but scores for different cell types were not combined for LINCS, unlike CMap.

2.2 Multi-step shortlisting of potential PINK1/PARK2-boosting compounds

In search of suitable drugs with a potential to be repurposed into treatment for PD, four steps of shortlisting were conducted as follows.

First step: gathering all compounds that upregulate PINK1/PARK2

Lists of drug-like compounds were obtained from the two databases, CMap and LINCS via the SPIED3 platform. Drugs predicted to down-regulate PINK1 and PARK2, indicated by the negative scores on both databases, were first eliminated.

Second step: shortlisting of PINK1/PARK2-upregulating compounds according to selection criteria

Compounds with positive scores in CMap and LINCS were shortlisted in the following order of preference until ~30 compounds were reached:

1. co-upregulation with higher score for PARK2 upregulation
2. co-upregulation with higher score for PINK1 upregulation
3. mono-upregulation of PARK2

Only the 30 top-ranking compounds were considered from both databases, out of the ($\leq$)250 drugs with positive scores (250 with negative scores), because results are capped at a maximum of 500 outputs per query on SPIED platform. CMap was preferentially utilised over LINCS because Z scores of the same drug on different cells types have not been not combined in LINCS, so CMap reflects drug activity relatively more accurately.

After compiling a list of drug compounds that co-upregulate PINK1 and PARK2 to varying degrees, the top 20 ranking compounds that are predicted to only upregulate PARK2 on CMap were added to the list, as the cutoff line has been set at CMap score $>$ 0.46 (Supplementary Table 3). This number was derived from the relative PARK2 regulation rank score of chlorcyclizine, the second and last compound from the combined profile on CMap that co-upregulate both PINK1 and PARK2, amongst the top 30 compounds (Supplementary Table 1).

Third step: shortlisting for suitability as PD treatment

The shortlisted drugs from the second step were then subject to literature-based scrutiny to rule out drugs that show the following limitations:

1. known emergence of toxicity or undesirable adverse effects following chronic dosing
2. low blood-brain barrier (BBB) penetration probability, indicated by log P $<$ 0

3. anticipated contraindications for use in PD

Fourth step: elimination of non-clinically used candidates

After the third step, drugs were further scrutinised according to their current clinical utility. Non-clinically used drugs were eliminated and only established drugs were taken forward for in vitro experiments, to assess whether their predicted upregulation of PINK1 and/or PARK2 would result in corresponding changes in the production of the respective proteins, PINK1 and Parkin.

2.3 In vitro assessment of drug effect on PINK1 and Parkin production in MCF-7 cells

MCF-7 human breast carcinoma cells (Sigma, 86012803) were used plated on T75 flasks for expanding, and on T25 flasks to be subsequently exposed to the three drug finalists and their respective vehicle controls for 24 hours to assess the protein production in vitro following drug treatment.

Culture medium used was Dulbecco's Modified Eagle Medium (DMEM)-glutaMAX (Thermo, 61965-026) supplemented with 1% penicillin-streptomycin (P/S) (Gibco, 15140122) and 10% foetal bovine serum (FBS) (Sigma, F2442). Plating and passaging was conducted with the regular trypsin method: First, cells were incubated in warmed 0.25% trypsin (Gibco, 15400-054) (1 ml for a T75 flask) at 37 ℃ for 5 minutes after three washes with phosphate-buffered saline (PBS). Then, the medium was added to dilute trypsin 1:10 (9 ml medium + 1 ml trypsin), followed by centrifugation at 1000 rpm for 5 minutes. Next, the supernatant was aspirated, and the cell pellet was resuspended evenly with 1 ml warmed medium. For expansion, cells were plated at a final density of 500,00 cells/10 ml medium for a T75; for drug treatment, cells were plated at 1500000 cells/3 ml for a T25 flask.

For drug treatment, the next day after plating on T25s, drugs previously dissolved in their corresponding vehicles were formulated in warmed FBS-containing DMEM-glutaMAX medium, so that the final drug concentration would be 10 µM (2.7 ml medium + 300 µl of 100 µM for a T25 flask). This concentration was chosen for consistency with that used for the transcriptional profiling. Then, after aspirating the old medium followed by three PBS washes, the drug-containing medium was added to the cultures. Cultures were exposed each drug and its individual vehicle controls for 24 hours before collection and lysis of

MCF-7 cells. Each drug and respective vehicle was tested on 6 independent cultures (n=6). See Table 1 for the companies from which drugs were obtained, and for their corresponding vehicle used as controls.

Compound	Company	Catalogue no.	Vehicle
Ethotoin	United States Pharmacopeia (USP)	1264501	0.4% DMSO
Oxybutynin	Sigma	O2881	0.1% ethanol
Sirolimus	Sigma	R0395	dH_2O

Table 1: Final drug candidates and vehicle controls All drugs were initially dissolved in their corresponding vehicles and further diluted in FBS-containing DMEM-glutaMAX medium that the final drug concentration would be 10 µM.

Cells were detached using the trypsin method as well, except that FBS-free medium was used to neurtalise the trypsin, and lysed by freeze-thawing and high-frequency sonication in lysis buffer. Proteins were frozen at -80 ℃ for storage purpose. After centrifugation (1000 rpm, 5 minutes) at 4 ℃, a bicinchoninic acid (BCA) analysis was carried out using the standard BCA assay kit (Thermo, 23225) to obtain protein concentrations of sample lysates by comparing against a bovine serum albumin (BSA) standard curve. They were subsequently diluted in dH_2O to 1 mg/ml protein to be used as western blotting samples for PINK1 and Parkin quantification.

2.4 Assessment of PINK1 and Parkin production in MCF-7 cells following drug treatment

Western blotting (WB) was carried out to determine protein expression of PINK1 and Parkin in MCF-7 in drug-treated MCF-7 cultures versus their vehicle controls.

Protein-containing lysates were mixed with the loading buffer (2X) at 1:1 and boiled at 95 ℃ for 5 - 10 minutes, followed by vortexing and cooling on ice. The Precision Plus Protein Dual Color Standards (BioRad, 1610374) was used as the molecular mass marker. Then, recombinant human proteins (PINK1 (Abcam, ab116177) and Parkin (Abcam, ab140806)) being the positive control and the boiled protein samples were loaded into 15% SDS-PAGE gels in duplicates. Proteins were separated across the gel in running buffer by applying 75 V for 2 hours and 10 minutes.

After gel electrophoresis, proteins were transferred to the nitrocellulose membrane in transfer buffer in a cassette run at 60 V for approximately 1.5 hours. The membrane was blocked by incubating it in 5% non-fat milk in PBST buffer for 1 hour at room temperature on a shaker. Then, it was incubated with primary antibodies, which were diluted in PBST-containing 2% milk (Marvel milk powder), overnight at 4 °C. The next day, two 5-minute washes in PBST were performed before incubated in secondary antibody also diluted in PBST-containing 2% milk for 1 hour at room temperature on a shaker. After the secondary antibody incubation, the membrane was washed twice in PBST for 5 minutes each and rinsed in 0.1M PBS before scanning with LI-COR's Odyssey CLx Near-Infrared Fluorescence Imaging System. Mock imaged blots were analysed using Image J/FIJI for densitometry data. See Table 2 for antibody details.

Antibody	Dilution	Species	Company	Catalogue no.
anti-Parkin (1°)	1:2000	mouse	Abcam	ab77924
anti-PINK1 (1°)	1:1000	rabbit	Abcam	ab23707
anti-GAPDH (1°)	1:10000	rabbit	Abcam	ab181602
LI-COR IR Dye 680nm anti-mouse (2°)	1:10000	goat	LI-COR	926-68070
LI-COR IR Dye 800nm anti-rabbit (2°)	1:10000	goat	LI-COR	926-32211

Table 2: Primary (1°) and secondary (2°) antibodies used in western blotting All antibodies were diluted in PBST-containing 2% milk.

Imaged blots were analysed with ImageJ to generate densitomtry data, followed by normalisation to the expression of housekeeping gene, glyceraldehyde 3-phosphate dehydrogenase (GAPDH).

2.5 Statistical analysis

Two-tailed unpaired student's t-test was used for analysis of densitometry data from western blots to compare between drug treatment and individual vehicle controls. All statistical analyses were performed using GraphPad Prism. P values smaller than 0.05 were considered statistically significant. All data are expressed as mean ± SEM.

3 Results

3.1 Shortlisting of PINK1/PARK2-boosting compounds

Utilising the two transcriptional databases (CMap and LINCS) on the SPIED3 platform, a four-step shortlisting process was carried out to determine compounds with a potential to be repositioned as PD treatment by upregulating PINK1/PARK2 and increasing PINK1/Parkin production, to promote the PINK1/Parkin pathway.

After the first and second shortlisting steps, primarily based on the scores reflective of PINK1 and/or PARK2 upregulation, a total of 33 compounds qualified (Fig 2): 2 and 10 compounds predicted to show co-upregulation from the combined profile of drug regulation on PINK1 and PARK2 expression from CMap and LINCS respectively (Supplementary Fig 1, 2), 3 extra top-ranking compounds predicted to only upregulate PARK2 from LINCS (Supplementary Fig 2), and 18 compounds predicted to only show PARK2 upregulation from CMap (Supplementary Fig 3).

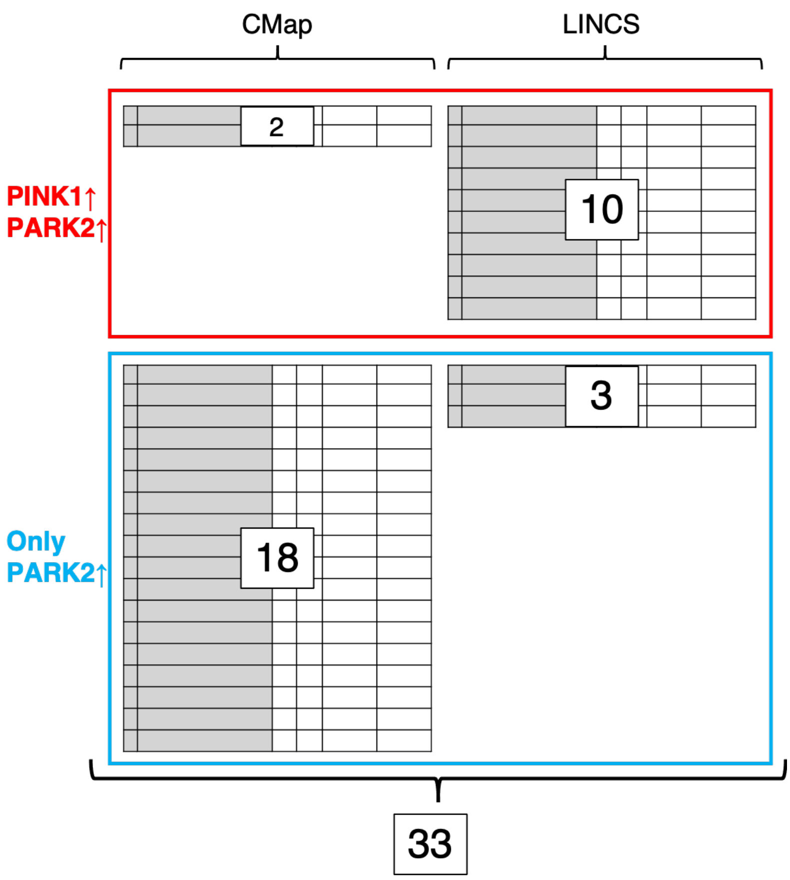

Figure 2. Step 2 shortlisting process Compiling the list from the seccond shortlisting step to be subject to literature-based scrutiny in the next (third) step to to rule out compounds unsuitable for treating PD.

These 33 compounds were further scrutinised in the third step to eliminate drugs that show characteristics rendering them unsuitable as PD treatment. The information gathered was compiled onto a list, with the corresponding colours showing the specific reasons for elimination in this shortlisting step (Fig 3, 4).

As a result of ruling out compounds unsuitable for treating PD, a total of 24 compounds were dropped from the 33 shortlist from the second step. This intermediate list of 9 candidates was further shortened by removing 6 compounds that are currently not in clinical use, eventually leading to the 3 drug finalists – ethotoin, oxybutynin, and sirolimus.

	Shortlisted list after step 2: 12 compounds			Step 3: 2 compounds	Step 4: 1 compound
	Compound	Gene regulation scores	Rank		
CMap (2)	ellipticine	PINK1(0.82), PARK2(0.51)	1 (combined profile) 13 (PARK2 only profile)	anti-cancer/cytotoxic	
	chlorcyclizine	PINK1(0.49), PARK2(0.46)	2 (combined profile) 18 (PARK2 only profile)	✓	up to Phase I
	Compound (Cell line)	Gene regulation scores	Rank on combined profile		
LINCS (10)	AT-13387 (YAPC)	PINK1(3.16), PARK2(2.51)	4	anti-cancer/cytotoxic	
	5-iodotubercidin (A549)	PARK2(3.00), PINK1(2.26)	9	anti-cancer/cytotoxic	
	BIIB-021 (YAPC)	PINK1(2.72), PARK2(2.46)	10	anti-cancer/cytotoxic	
	NVP-TAE226 (HT29)	PINK1(2.72), PARK2(2.35)	13	anti-cancer/cytotoxic	
	fulvestrant (MCF7)	PINK1(2.59), PARK2(2.24)	16	anti-cancer/cytotoxic	
	HG-13-10-04 (HT29)	PINK1(2.52), PARK2(2.27)	18	no information found	
	ganetespib (YAPC)	PINK1(2.59), PARK2(2.09)	22	anti-cancer/cytotoxic	
	sirolimus (HCC515)	PARK2(2.35), PINK1(2.27)	25	✓	✓
	metronidazole (MCF7)	PINK1(2.38), PARK2(2.14)	26	antibiotic	
	GDC-0941(BT20)	PARK2(2.24), PINK1(2.21)	29	anti-cancer/cytotoxic	

Uncoloured ticked boxes – qualified ✓

Black boxes – already eliminated

Coloured boxes – reasons for elimination
- Undesirable adverse effects with chronic use
- Low BBB penetration (negative log P values)
- Anticipated contraindications for use in PD
- No relevant information found
- Not currently in clinical use

Overall	Shortlisted list after step 2: 12 + 21 = 33 compounds	Step 3 outcome: 2 + 7 = 9 compounds	Step 4 outcome: 1 + 2 = 3 compounds

Figure 3. Shortlisting of compounds showing co-upregulation of PINK1 and PARK2 Amongst the 33 compounds shortlisted after second step, 12 compounds were predicted to show co-upregulation from both CMap and LINCS (Supplementary Fig 1, 2). 10 and 1 compounds were dropped from further analysis after the third and the fourth steps respectively, because of various reasons, indicated by different colours: yellow - unsuitable for chronic administration; purple - low probability of BBB penetration; orange - anticipated contraindications for use in PD; grey - no relevant information found; green - no current clinical utility. This resulted in only one compound that co-upregulates PINK1 and PARK2, sirolimus, qualifying as one of the 3 drug candidates shortlisted after the final step.

	Shortlisted list after step 2: 21 compounds			Step 3: 7 compounds	Step 4: 2 compounds
	Compound	Gene regulation score	Rank of PARK2 score		
Cmap (18)	skimmianine	PARK2(0.64)	1	✓	preclinical
	apigenin	PARK2(0.63)	2	✓	preclinical
	hexestrol	PARK2(0.62)	3		
	thioguanosine	PARK2(0.60)	4		
	sitosterol	PARK2(0.60)	5		
	pyridoxine	PARK2(0.58)	6	negative log P (-0.95)	
	ethotoin	PARK2(0.58)	7	✓	✓
	0175029-0000	PARK2(0.58)	8		
	oxybutynin	PARK2(0.57)	9	✓	✓
	fluorometholone	PARK2(0.55)	10		
	fusaric acid	PARK2(0.53)	11	✓	preclinical
	ampicillin	PARK2(0.52)	12	antibiotic	
	roxithromycin	PARK2(0.50)	14	antibiotic	
	clemizole	PARK2(0.47)	15		
	aminophenazone	PARK2(0.46)	16		
	harmine	PARK2(0.46)	17	✓	preclinical
	theobromine	PARK2(0.46)	19	negative log P (-0.77)	
	remoxipride	PARK2(0.46)	20	antipsychotic	
	Compound (Cell line)	Gene regulation score	Rank on combined profile		
LINCS (3)	troglitazone (MCF7)	PARK2(8.37)	1	✓	approved and withdrawn
	AT-7519 (PC3)	PARK2(5.82)	2	anti-cancer/cytotoxic	
	mitoxantrone (NPC.TAK)	PARK2(5.70)	3	anti-cancer/cytotoxic	

Uncoloured ticked boxes – qualified ✓

Black boxes – already eliminated

Coloured boxes – reasons for elimination:
- Undesirable adverse effects with chronic use
- Low BBB penetration (negative log P values)
- Anticipated contraindications for use in PD
- No relevant information found
- Not currently in clinical use

Overall	Shortlisted list after step 2: 12 + 21 = 33 compounds	Step 3 outcome: 2 + 7 = 9 compounds	Step 4 outcome: 1 + 2 = 3 compounds

Figure 4. Shortlisting of PARK2-boosting compounds Amongst the 33 compounds shortlisted after second step, 21 compounds were predicted to only upregulate PARK2 from both CMap and LINCS (Supplementary Fig 1, 2). 14 and 5 compounds were dropped from further analysis after the third and the fourth steps respectively, because of various reasons, indicated by different colours: yellow - unsuitable for chronic administration; purple - low probability of BBB penetration; orange - anticipated contraindications for use in PD; grey - no relevant information found; green - no current clinical utility. This resulted in two PARK2 only-upregulating compounds, ethotoin and oxybutynin, qualifying as the other 2 candidates out of the 3 shortlisted after the final step.

Intermediate list of potential compounds

Further literature-based scrutiny of shortlisted compounds after the third step led to the intermediate list of 9 compounds that not only lack the pre-assigned limitations being the elimination criteria in step 3, but also showed potential to be PD treatment apart from their upregulation on PINK1 and/or PARK2. Although there were only 3 out of 9 compounds qualified ultimately, they may be of interest for further investigation nonetheless. Additional information of these potential compounds has been tabulated in the following (Table 3).

Compound	Additional information
chlorcyclizine	anti-histamine
sirolimus (HCC515)	immunosuppressant, primarily used after renal transplantation
skimmianine	analgesic, antispastic, sedative, antiplatelet, antidepressant, anti-inflammatory agent
apigenin	antioxidative, anti-inflammatory, and neuroprotective agent
ethotoin	antiepileptic, anticonvulsant
oxybutynin	anti-cholinergic, currently used to treat overactive bladder
fusaric acid	increase dopamine levels by inhibiting dopamine beta-hydroxylase, the enzyme that converts dopamine to norepinephrine
harmine	anti-inflammatory, treat TBI (traumatic brain injury)
troglitazone (MCF7)	anti-diabetic, withdrawn in 1997 in the US as cardiovascular risks and hepatotoxicity outweigh benefits to type 2 diabetes

Table 3: Information for intermediate list of 9 potential compounds after step 3 Information of compounds that passed the third step of shortlisting, comprising the intermediate list of 9 compounds. A range of information of potential interest have been included such as proposed mechanisms of action, potential therapeutic effects on other pathways helpful for treating PD, and potential side effects.

Drug finalists

Three final drugs were resulted from the four-step shortlisting process: ethotoin and oxybutynin from CMap, and sirolimus from LINCS. Table 4 summarises the noteworthy characteristics of the three drugs.

Compound	Rank	PINK1	PARK2	Log P	Class
Ethotoin	CMap: 7 (Supplementary Fig 3)	\	upregulate	1.05	anticonvulsant
Oxybutynin	CMap: 9 (Supplementary Fig 3)	\	upregulate	4.3	anticholinergic
Sirolimus	LINCS: 25 (Supplementary Fig 2)	upregulate	upregulate	4.3	immunosuppressant

Table 4: Drug information of 3 final candidates The 3 drug finalists' ranks from their respective databases, their predicted regulatory effects on PINK1 and PARK2, their BBB penetration probability indicated by the log P value, and the drug class.

3.2 Drug effects on PINK1 and Parkin production

Next, western blotting was carried out to determine whether the upregulation of PINK1/PARK2 corresponded to increased protein expression of PINK1/Parkin in drug-treated MCF-7 cultures.

Ethotoin

Parkin expression was significantly higher with drug treatment (1.10 $\pm$ 0.03) than without it (0.41 $\pm$ 0.05) (P<0.0001), showing a 258% increase. PINK1 expression was not significantly increased by drug treatment compared to its vehicle control (drug: 0.27 $\pm$ 0.03; vehicle: 0.30 $\pm$ 0.02) (P=0.5300) (Fig 5B).

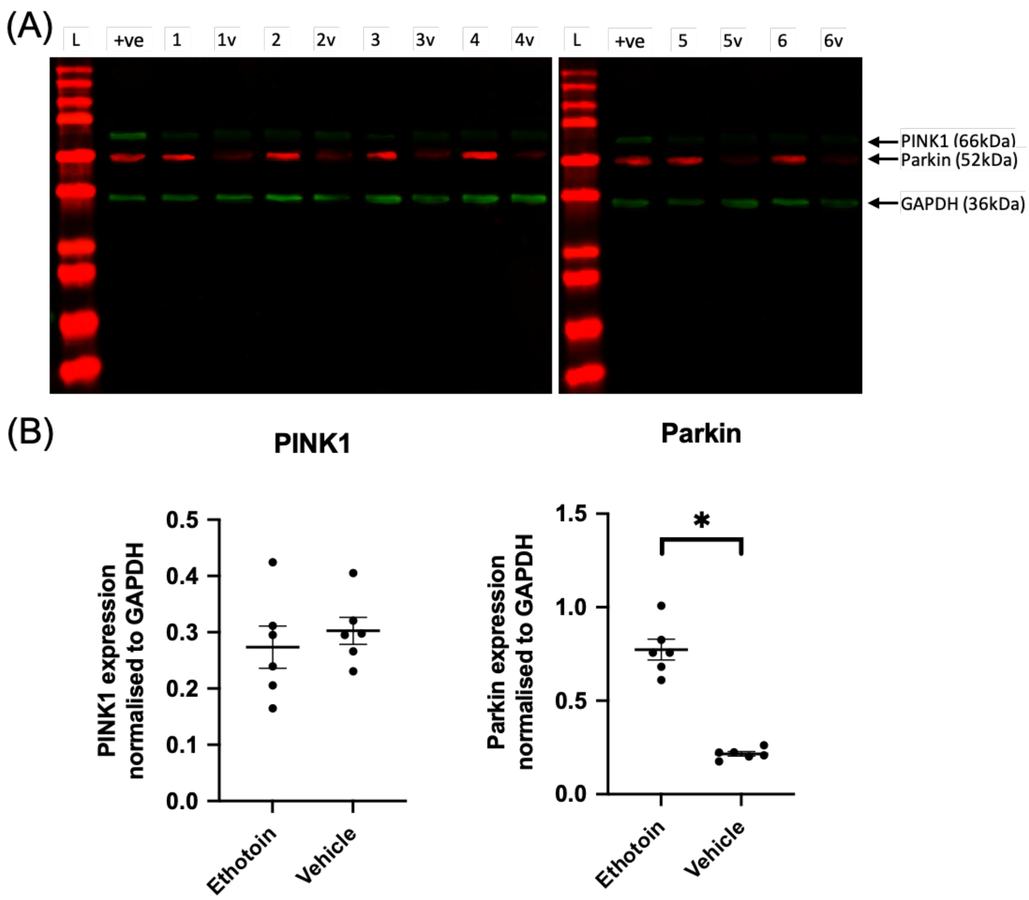

Figure 5. Effects of ethotoin treatment on PINK1 and Parkin expression MCF-7 cells after 24-hour exposure to ethotoin (n=6). (A) Imaged full western blots showing PINK1 (66 kDa), Parkin (52 kDa), and GAPDH (36 kDa) bands. L - ladder; +ve - positive control; 1 - sample 1 treated with drug, 1v - sample 1 treated with vehicle, and so on. Samples 5 and 6 were run on a separate gel with equivalent positive control. (B) Densitometry data showing relative expression of PINK1 and Parkin normalised to GAPDH expression. Parkin expression: P<0.0001; PINK1 expression: P=0.5300. Each data point corresponds to each sample. Unpaired student's t-test was used for analyses. * indicates statistical significant difference at the 0.05 level. Data are expressed as mean $\pm$ SEM.

Oxybutynin

Parkin expression was significantly higher with drug treatment (1.27 ± 0.18) than without it (0.29 ± 0.02) (P=0.0003), showing a more than three-fold increase (+332%). PINK1 expression was not significantly increased by drug treatment compared to its vehicle control (drug: 0.26 ± 0.01; vehicle: 0.23 ± 0.03) (P=0.3846) (Fig 6B).

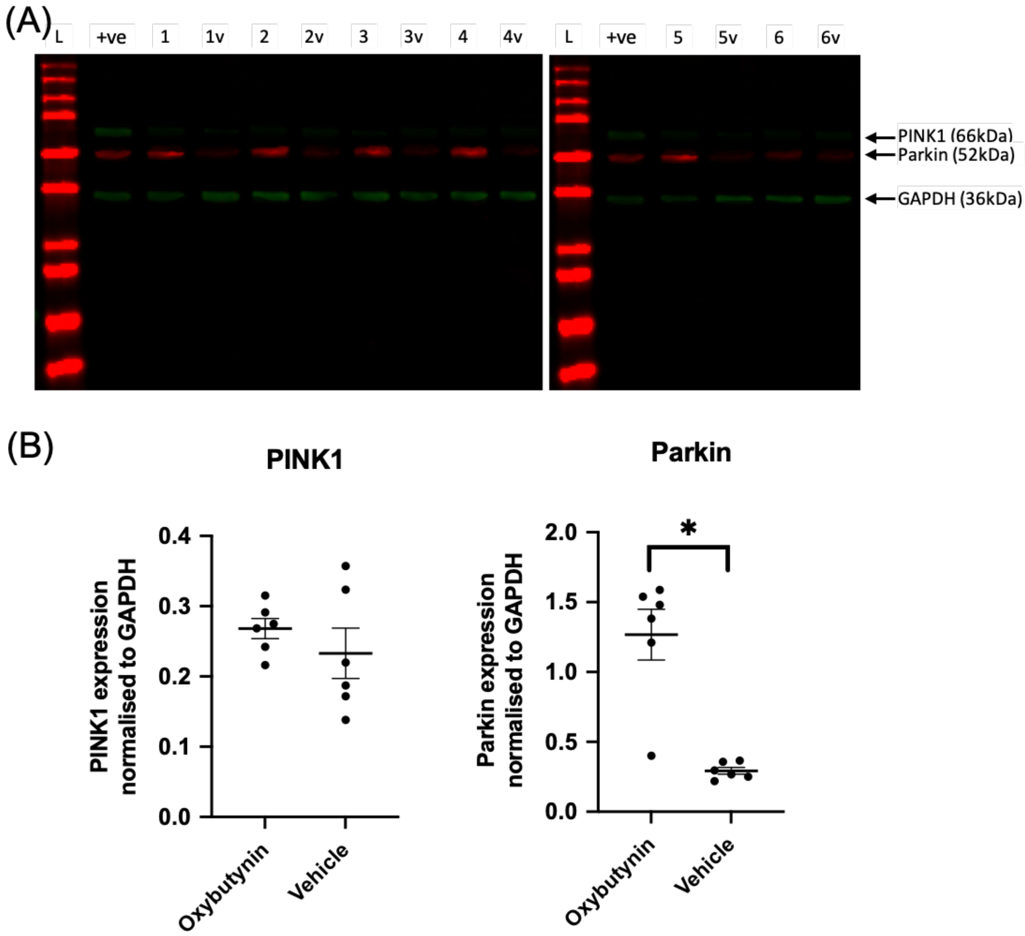

Figure 6. Effects of oxybutynin treatment on PINK1 and Parkin expression MCF-7 cells after 24-hour exposure to oxybutynin (n=6). (A) Imaged full western blots showing PINK1 (66 kDa), Parkin (52 kDa), and GAPDH (36 kDa) bands. L - ladder; +ve - positive control; 1 - sample 1 treated with drug, 1v - sample 1 treated with vehicle, and so on. Samples 5 and 6 were run on a separate gel with equivalent positive control. (B) Densitometry data showing relative expression of PINK1 and Parkin normalised to GAPDH expression. Parkin expression: P=0.0003; PINK1 expression: P=0.3846. Each data point corresponds to each sample. Unpaired student's t-test was used for analyses. * indicates statistical significant difference at the 0.05 level. Data are expressed as mean ± SEM.

Sirolimus

Parkin expression was significantly higher with drug treatment (0.77 ± 0.06) than without it (0.21 ± 0.01) (P<0.0001), although showing the smallest percentage increase (+167%) compared to the other two drugs. Also unlike ethotoin and oxybutynin, PINK1 expression only showed an increased following sirolimus treatment by 427% (1.03 ± 0.03) compared to its vehicle control (0.20 ± 0.04) (P<0.0001) (Fig 7B).

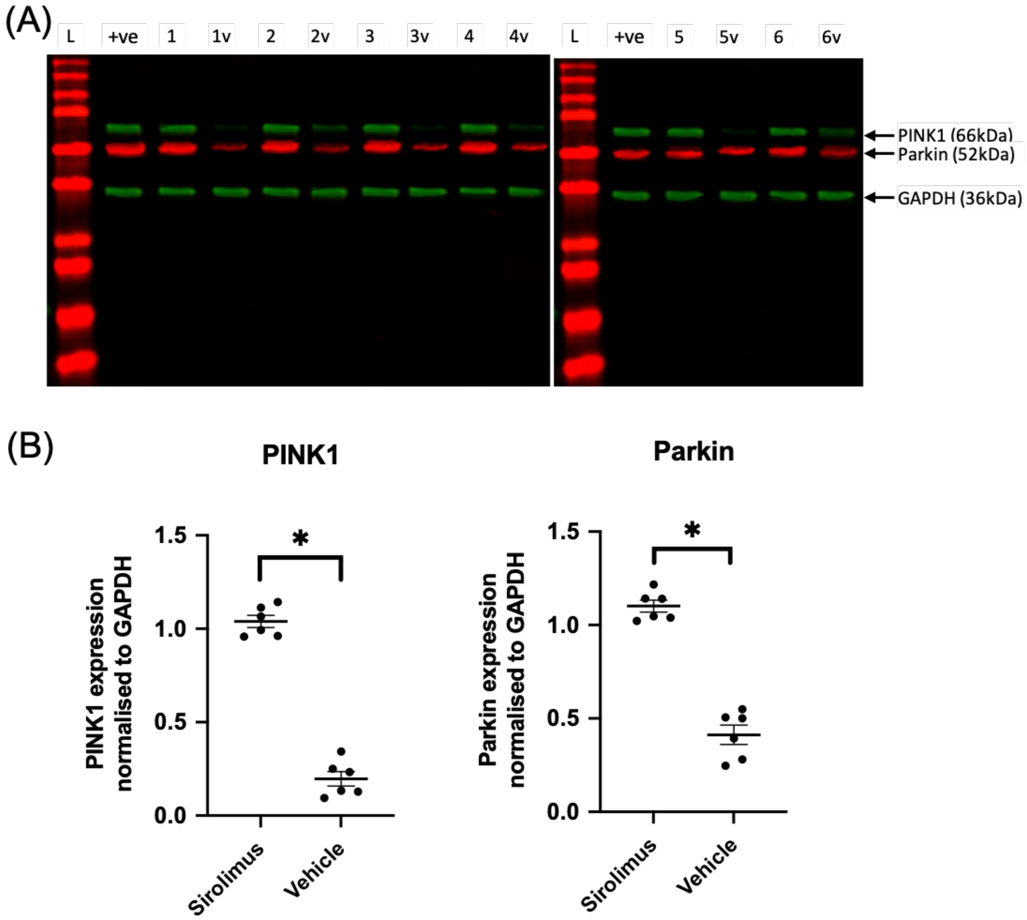

Figure 7. Effects of sirolimus treatment on PINK1 and Parkin expression MCF-7 cells after 24-hour exposure to sirolimus (n=6). (A) Imaged full western blots showing PINK1 (66 kDa), Parkin (52 kDa), and GAPDH (36 kDa) bands. L - ladder; +ve - positive control; 1 - sample 1 treated with drug, 1v - sample 1 treated with vehicle, and so on. Samples 5 and 6 were run on a separate gel with equivalent positive control. (B) Densitometry data showing relative expression of PINK1 and Parkin normalised to GAPDH expression. Parkin expression: P<0.0001; PINK1 expression: P<0.0001 Each data point corresponds to each sample. Unpaired student's t-test was used for analyses. * indicates statistical significant difference at the 0.05 level. Data are expressed as mean ± SEM.

	Change in expression (compared to vehicle)	
Compound	**PINK1**	**Parkin**
Ethotoin	\	+258%
Oxybutynin	\	+332%
Sirolimus	+427%	+167%

Table 5: Effects of drug finalists on PINK1 and Parkin expression Summary table showing changes in PINK1 and Parkin production in MCF-7 cells following treatment of the three drug finalists versus their vehicle controls.

Overall, expression levels of Parkin were significantly higher in samples with all three drugs compared to each of individual vehicle controls. PINK1 expression was only promoted by sirolimus treatment. PINK1 was not significantly expressed higher after ethotoin and oxybutynin treatments compared to their individual vehicle controls.

4 Discussion

The targeted repositioning strategy was conducted in a step-wise manner, starting with the in silico screening to interrogate the CMap and LINCS transcriptional databases accessed via the the SPIED3 tool (Williams, 2013). Following literature-based mining in multiple steps of shortlisting, potential drug-like compounds that were identified to be unsuitable for use were removed: the first step removed all compounds that did not upregulate PINK1 and/or PARK2; the second step created a shortlist of 33 PINK1/PARK2-upregulating compounds from top ranking compounds on both databases to be subject to further analyses; the third step removed compounds that were known to present toxicity following chronic use, not cross the BBB or be contraindicated for use in people with PD, leading to an intermediate list of 9 remaining drugs of interest; the fourth step selected only currently established drugs in clinical use. Finally, three drugs were obtained – ethotoin, oxybutynin, and sirolimus.

These finalists were screened for the ability to boost PINK1 and Parkin protein production in vitro. Although the MCF-7 cell line bears no resemblance to neural cells relevant in PD, functionally or in terms of their cell lineages, their use was deliberate because MCF-7 cells had been used to populate the CMap transcriptional database, which was preferentially utilised over LINCS in this study, generating confidence in the translational potential between the gene regulation and the accomplished changes in protein production.

4.1 Drug finalists

All three drugs showed the predicted increase in PINK1 and/or Parkin production in vitro, indicating the translational potential between the predicted PINK1/PARK2 upregulation and the target respective changes in protein production. Amongst the three, oxybutynin showed the highest increase in Parkin production (+332%), but sirolimus showed increase in both expression of PINK1 (+427%) and Parkin (+167%) (Table 5), as was predicted to co-upregulate both PINK1 and PARK2.

Overall, all three drugs should be taken forward for further studies. But of the three, ethotoin is primarily favoured, the reasons for which are as follows.

Initially, it was pre-assigned in the criteria for the second shortlisting step that drugs showing PINK1/PARK2 co-upregulation would be preferred over those only showing PARK2 upregulation. This is derived from the knowledge of PINK1 and Parkin acting in the same mitochondrial pathway (Kazlauskaite & Muqit, 2015; Lee et al., 2017), thus hypothesised that increased production of both proteins would show a synergistic effect to improve MQC where it may not be optimal in PD. However, with closer inspection of the pathway and the specific roles played by PINK1 and Parkin, it is possible that the elevated expression is only required for Parkin and not PINK1 for the beneficial effects for the following two reasons. Firstly, Parkin acts downstream of PINK1, catalysing the ubiquitination of downstream substrates including PARIS to produce neuroprotective effects particularly during stress (Lee et al., 2017; Shin et al., 2011). Parkin overexpression was also known to mask and rescue KO-PINK1 phenotype (Kazlauskaite & Muqit, 2015). Secondly, PINK1, as the sensor of mitochondrial damage, propagates the signal of damage or stress by its accumulation in the OMM. Increased expression of functional PINK1 protein would not necessarily show beneficial effects in the pathway, when the mechanism of signalling of stress involves the dynamic changes in PINK1 levels itself, unless the crux of an impaired PINK1/Parkin pathway in PD lies in the lack of accumulation of functional PINK1 for effective transmission of stress signals. This is an outstanding question that warrants future investigation, given the relevance PINK1 loss in PD as one of the most loss-of-function common mutations in autosomal recessive fPD (Chakraborty et al., 2020; Ge et al., 2020; Khatri et al., 2020).

On this basis, sirolimus is the least favoured finalist amongst the three, as it shows the smallest percentage increase in Parkin expression, whereas oxybutynin with the largest increase in Parkin production would be the most favoured (Table 5). Nonetheless, oxybutynin may not be as suitable a fit to treat PD as initially thought to be, as an anticholinergic agent. Although anticholinergic drugs were the earliest pharmacological agents used in the treatment of PD, such as procyclidine and biperiden, (Brocks, 1999), the chronic use of specific types of anticholinergic medication has been associated with the development of dementia (Coupland et al., 2019). Hence, it has been advised that anticholinergic drugs should be prescribed with caution in middle-aged and older adults (Coupland et al., 2019), corresponding to the age group more vulnerable to developing PD in general, being an age-related neurodegenerative diseases (Rouaud et al., 2020). This finding from further literature research suggests that the long-term use of oxybutynin may result in cognitive worsening as a side effect, not only amplifying symptoms from cognitive impairments, which is common though not universal in PD, but also generating problems de novo arising from declined cognitive functions for PD patients that did not have them in the first place. The use of anticholinergic drugs puts all PD patients at a risk of potential deterioration of cognitive abilities, particularly for patients already living with PD-related dementia (PDD) (Brandão et al., 2020).

With regards to side effects of the drugs, sirolimus may also be not the most desirable option, on top of its smallest percentage increase in Parkin expression. Clinically, it is the primary immunosuppressant used after renal transplants (Liu et al., 2016). Adverse effects related to immunosuppression do not directly coincide with PD-specific side effects, also the reason for it being included after the third shortlisting step. Nevertheless, the use of immunosuppressants may not be most desirable in the treatment of PD, given the proneness to falls in PD patients with increased age, and having postural instability as a primary motor symptom (Chakraborty et al., 2020). In fact, the two most prevalent causes of death in sPD were pneumonia (30%), and sudden collapse (10%) (Pennington et al., 2010). Therefore, administering sirolimus to PD patients may indirectly increase mortality risk due to increased risks of infections and immunosuppression.

On the whole, ethotoin appears to be the most promising drug finalist. It is a hydantoin derivative used as an anticonvulsant. It exerts an antiepileptic effect without causing general central nervous system depression. The mechanism of action is not completely known, but has been proposed to raise the normal seizure threshold, and to abolish the primary focus of seizure discharges (Wishart et al., 2006). Although its proposed mechanism of action does not directly counteract proposed pathophysiology and pathogenesis in PD, by stabilising neuronal membranes thereby preventing the spread of seizure activity at the motor cortex, ethotoin may also show extra symptom-relieving effects by its potential reduction of tremors in PD, on top of its increase in Parkin production.

Although ethotoin showed the most potential, all three drugs should be taken forward for possible future experiments, including the assessment of drug-induced PINK1 and Parkin production in neuronal-glial co-cultures. In addition to the in vitro system, protein production could also be examined in vivo in rat striatum following drug dosing. These would further increase the confidence in translational potential between the gene regulatory effects of the drugs and the target changes in the production of proteins. qRT-PCR could be a further confirmatory step directly measuring PINK1 and PARK2 mRNA levels to confirm that the observed changes in protein production were resulted from transcriptional changes. Additionally, the neuroprotective potential of the drugs could be tested in animal models, such as the 6-hydroxydopamine(OHDA)-lesioned rat models of PD, by examining how well the promoted PINK1/Parkin pathway following dosing protects against 6-OHDA-induced nigral cell loss in rats.

4.2 Justification of repositioning strategy

The following subsection discusses the detailed justification of the methods and different criteria involved in the multi-step shortlisting process.

CMap and LINCS transcriptional databases

As described in the Materials and Methods section, the scores indicating the gene regulation effects of the same drug on different cells types have not been not combined for LINCS, whereas they have been combined for CMap, so CMap reflects drug activity relatively more accurately. This explains the preferential use of CMap over LINCS for profile only showing compounds predicted to upregulate PARK2. For instance, on the combined profile on LINCS, the same drug, mitoxantrone, has appeared twice amongst the top 30 outputs with two different scores. Ranked in 3rd place, PARK2 upregulation by mitoxantrone on NPC.TAK cell line showed a score of 5.70, in contrast to the 20th-ranking score of 4.76 indicating the same upregulation by the same compound on the NPC cell line (Supplementary Fig 2). This is undesirable as it was expected that the top 30 query outputs should correspond to the top 30 compounds that upregulate PARK2 and/or PINK1 to the highest degrees, explaining the primary utilisation of CMap over LINCS.

The term "combined profile" was used to describe profiles of outputs from a query of more than one genes. On combined profiles, compounds are ranked according to the sums of scores of all the genes the compound is predicted to regulate. Supplementary Figures 1 and 2 show the combined PINK1/PARK2 profiles extracted from CMap and LINCS respectively. Although the reality of drug-gene interaction is not reflected from the simple score addition, combined profiles were used to obtain compounds that (up)regulate both genes. For example, it would not have been intuitive to users of SPIED3 that ellipticine and chlorcyclizine actually upregulate both PARK2 and PINK1, if only the PARK2 profile had been consulted, because with the PARK2 score alone, the CMap ranks of ellipticine and chlorcyclizine were not amongst the top 10 drugs, 13 and 18 respectively (Supplementary Fig 1).

Criteria in shortlisting steps

The first step gathered all the compounds with positive scores from both CMap and LINCS, indicating PINK1/PARK2 upregulation. In the second step, amongst all the compounds that co-upregulate PINK1 and PARK2, those that show a higher degree of PARK2 upregulation were favoured over those showing a higher degree of PINK1 upregulation. As PINK1 is upstream of Parkin in the pathway (Kazlauskaite & Muqit, 2015; Lee et al., 2017), it was hence speculated that neuroprotection is primarily resulted from overexpression of Parkin rather than PINK1. Simply put, this means for compounds upregulating both genes, it is more significant that the score for PARK2 is higher than that for PINK1 than vice versa. Also, compounds that only upregulate PINK1 expression was thoroughly not considered in this step, due to the speculated smaller role compared to Parkin, as discussed in the last subsection, evidenced by the phenomenon of PARK2 overexpression masking KO-PINK1 phenotype and the inability of PINK1 overexpression to rescue KO-PARK2 phenotype (Kazlauskaite & Muqit, 2015).

After the third step, drugs that are unsuitable were removed for various reasons (Fig 3, 4). Firstly, drugs with anticipated unwanted side effects following long-term dosing were removed; this include antibiotic and anti-cancer drugs. Prolonged antibiotic use was not desirable due to issues with the emergence of antibiotic-resistant bacteria, whereas cytotoxic anti-cancer agents were avoided because they are inherently "toxic to cells", in addition to the resultant cancer-related cognitive impairment and fatigue with which PD patients already struggle with. Secondly, drugs with negative log P values were removed, as it indicates the inability of the compound to penetrate the BBB. Eliminated compounds include pyridoxine and theobromine which had log P values of -0.95 and -0.77 respectively (Fig 4). Thirdly, drugs with known counteractive mechanisms of action were eliminated. For example, remoxipride (Fig 4), an antipsychotic, was removed, as it treats schizophrenia by selectively inhibiting dopaminergic activity, which is already progressively lost over the course of disease progression in PD patients, in addition to its predicted adverse effects coinciding with characteristic PD symptoms, such as tremor and rigidity (Laursen & Gerlach, 1986). Finally, for the rest of the drugs that were not on the 9-compound intermediate list entering the final (fourth) shortlisting step, there was either very little or no information about the compounds themselves and about their possibility of showing contraindications for use in PD, and was therefore removed from the intermediate list.

In the fourth step, only drugs with current clinical utility were selected for the subsequent in vitro assessment in MCF-7 cells. This is to increase the probability of successful development of a repurposed drug, because this step eliminates preclinical or Phase I drugs that have not passed Phase II clinical trials, which do bear the chance of being withdrawn due to common types of toxicity such as cardiotoxcity, in contrast to clinically used drugs that have already undergone safety and toxicity tests. These tests were also conducted with a large pool and wide range of human volunteers, on whom the established drugs had already been tested previously in Phase III trials, and have been proven to be safe to use at a certain dose range with an established administration route. For instance, a Phase I drug (drug A) and an established drug (drug B) both showed potential to be repositioned as treatment for PD. Both of them would be required to undergo the clinical trial phases again because they were to be repurposed for treating PD, a different disease from the one they were originally developed to treat. This implies the need to re-trial each of their efficacies against currently established PD drugs. In this scenario, drug B would stand a greater chance of success objectively, because failing safety tests would not be a concern for furthering the development of drug B, but might be one for drug A.

All in all, careful consideration has been included in the setting up of the criteria for each of the steps to ensure a high-quality and rigid shortlisting process to filter out compounds with the highest potentials for being repurposed to treat PD.

Shortlisted finalists

At first, compounds that show co-upregulation of PINK1 and PARK2 were originally preferentially selected in the second step of shortlisting, versus those only showing upregulation of one gene. However, eventually, amongst the final three candidates, only sirolimus that co-upregulate both genes was qualified, whilst ethotoin and oxybutynin were only predicted to upregulate PARK2. This mismatch between what was preferentially targeted and what was the resultant outcome is because of two reasons. Firstly, there were fewer drugs co-upregulating PINK1 and PARK2 compared to drugs that significantly upregulate only one of the two genes to start with. Secondly, most (10/12) drugs predicted to show co-upregulation were from LINCS, and interestingly, most compounds shown in LINCS query results ended up to be antibiotics or anti-cancer therapeutics after literature-based mining (Fig 3, 4), which make them unsuitable for chronic use. As a result, many compounds predicted to show co-upregulation did not successfully pass the third step of shortlisting. In fact, after the third step, there were only 2 compounds out of the 9 shortlisted that co-upregulate both PINK1 and PARK2: chlorcyclizine and sirolimus. Only sirolimus remained after the final step eliminating drugs that are not currently in clinical use, as chlorcyclizine has only passed Phase I trials (Fig 3). Moreover, as aforementioned, the degree of PARK2 upregulation ultimately outweights the need for both genes to be upregulated, owing to the more significant role of Parkin downstream of PINK1 in the pathway.

4.3 Perspectives regarding drug repurposing

The initial preference for drugs showing co-upregulation over mono-upregulation was partially refuted at a later stage of the project by the highlighted emphasis on the degree of PARK2 upregulation, corresponding to the magnitude of increase in Parkin production. One might pose the question of whether drug repositioning is an effective and more efficient strategy than the classical drug discovery and development system. In fact, some regard drug repurposing as a "well-intentioned but misguided strategy" (Tran & Prasad, 2020). Nonetheless, in this project, it could be argued that the emphasis on PARK2 upregulation was never neglected, as evidenced by the preferential selection of co-upregulating drugs with a higher score for PARK2 upregulation over those with a higher PINK1 upregulation score in the second shortlisting step. Hence it was only a matter reprioritisation.

Inspired from this project, the following three conditions are proposed as ideal conditions to be met in the consideration of applying the targeted repositioning strategy as an alternative to the conventional protracted and costly process of drug discovery. Firstly, there should be sufficient information of the target pathway of the disease in the existing literature, so that one can be confident in deducing that alterations in this pathway could potentially beneficial for treating the disease. Secondly, in identifying a molecular target in the pathway, the mechanism of deriving therapeutic benefits needs to be relevant to the increased or decreased expression of (a) protein(s). Thirdly, the disease of interest should involve alterations in a multitude of pathways; in this way, the aim for the shortlisting of drugs could be more precise, in terms of targeting drugs that not only show a high score in the target gene regulation, but also show potential in altering the other pathways in a therapeutically beneficial way. Moreover, targeting a disease with the involvement of multiple pathways means a higher chance of a "hit" of one of the pathways by the potential drugs to be repurposed.

All in all, it could be contended that this is a strategy worthy of attempting for diseases such as PD that meet all three of these conditions, and also currently lack cure.

Furthermore, it was also the use of this targeted repositioning strategy in this project that uncovered a number of drugs shortlisted in the third step on the intermediate list of 9 compounds (apart from the 3 finalists) with much potential for developing into PD treatment, although they did not qualify ultimately due to lack of current clinical utility (Table 3). For example, apigenin, an experimental drug, was found to be, anti-inflammatory and antioxidative, which match with the mechanisms of action of some current symptom-relieving PD drugs. It has even recently been shown to have neuroprotective effects in neuroinflammation models associated to Alzheimer's Disease (Dourado et al., 2020), implying significant potential in therapeutic effects in PD with neuroinflammation being a key contributer to the pathogenesis of PD. Another compound of interest, fusaric acid, also showed exciting potential as an inhibitor of dopamine beta-hydroxylase, thereby predicted to improve PD condition by increasing dopamine concentration, working similarly as most commonly known PD drug, levodopa. Further investigation into these drugs should be undertaken in light of their coincidental mechanisms of action producing potential therapeutic effects in PD, which is in addition to their upregulatory effects of PARK2, uncovered by the bioinformatics approach utilised in the targeted repurposing strategy in this project.

Finally, it should be mentioned that COVID-19 has been a huge disruption to the project due to the inability to access to lab facility, meaning the mock experimental results were not meaningful. Nevertheless, the significance of PINK1/Parkin pathway highlighted in this thesis, the proposed future experiments based on critical analysis of the current literature, and the potential in the novel strategy of targeted repositioning using bioinformatics tools are all valid. Further investigation into PINK1/Parkin pathway and trialling of the target respositioning strategy should be undertaken in the future, in hopes of uncovering drug candidates for treating PD, but also other progressive diseases with no cure at present.

5 Conclusion

This project presents an example of the use of targeted repositioning strategy, which has the potential to repurpose potential drugs for diseases currently without cure, such as PD.

In this project, 33 compounds initially gathered from the CMap and LINCS transcriptional databases were shortlisted over multiple steps into three final drugs – ethotoin, oxybutynin and sirolimus.

All three drugs showed the predicted changes in PINK1 and Parkin production when applied in vitro, corresponding to the respective PINK1/PARK2 upregulation predicted from the databases. Although ethotoin is the most promising, all three should be further analysed for neuroprotective potential in animal PD models, in the hope of moving another step closer to the discovery of a disease modifying drug treatment for PD.

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Appendices

	compound	skew	score	gene	gene
1	**ellipticine**	1.00	1.33	PINK1(0.82)	PARK2(0.51)
2	**chlorcyclizine**	1.00	0.95	PINK1(0.49)	PARK2(0.46)
3	quinostatin	1.00	0.93	PINK1(0.93)	
4	thiamine	1.00	0.89	PINK1(0.89)	
5	berberine	1.00	0.84	PINK1(0.84)	
6	DL-thiorphan	1.00	0.84	PINK1(0.84)	
7	esculetin	1.00	0.84	PINK1(0.84)	
8	5211181	1.00	0.81	PINK1(0.81)	
9	procarbazine	1.00	0.81	PINK1(0.81)	
10	halofantrine	1.00	0.80	PINK1(0.80)	
11	PHA-00846566E	1.00	0.80	PINK1(0.80)	
12	hycanthone	1.00	0.79	PINK1(0.79)	
13	folic_acid	1.00	0.79	PINK1(0.79)	
14	rottlerin	1.00	0.78	PINK1(0.78)	
15	convolamine	1.00	0.78	PINK1(0.78)	
16	butacaine	1.00	0.78	PINK1(0.78)	
17	hexetidine	1.00	0.77	PINK1(0.77)	
18	nimodipine	1.00	0.77	PINK1(0.77)	
19	mitoxantrone	1.00	0.76	PINK1(0.76)	
20	sulfamonomethoxine	1.00	0.75	PINK1(0.75)	
21	dropropizine	1.00	0.74	PINK1(0.74)	
22	methacholine_chloride	1.00	0.74	PINK1(0.74)	
23	metixene	1.00	0.73	PINK1(0.73)	
24	podophyllotoxin	1.00	0.73	PINK1(0.73)	
25	dihydroergotamine	1.00	0.72	PINK1(0.72)	
26	nortriptyline	1.00	0.72	PINK1(0.72)	
27	bromperidol	1.00	0.71	PINK1(0.71)	
28	bumetanide	1.00	0.71	PINK1(0.71)	
29	vigabatrin	1.00	0.71	PINK1(0.71)	
30	tetraethylenepentamine	1.00	0.70	PINK1(0.70)	

Supplementary Figure 1. Combined profile of drug regulation on PINK1 and PARK2 expression on CMap extracted from SPIED3 showing the top 30 compounds The outputs are ranked based on the expression levels across the 1309 drug-like compounds in the CMap database. The results of this query (PINK1, PARK2) was cropped to show only the 30 top ranking compounds. Only the compounds predicated to co-upregulate both PINK1 and PARK2 were subject to further literature-based scrutiny after the second step of shortlisting. Compounds in bold were taken forward into the third step.

	compound	skew	score	gene	gene
1	**troglitazone(MCF7)**	1.00	8.37	PARK2(8.37)	
2	**AT-7519(PC3)**	1.00	5.82	PARK2(5.82)	
3	**mitoxantrone(NPC.TAK)***	1.00	5.70	PARK2(5.70)	
4	**AT-13387(YAPC)**	1.00	5.67	PINK1(3.16)	PARK2(2.51)
5	WZ-3105(PC3)	1.00	5.62	PARK2(5.62)	
6	bardoxolone-methyl(HA1E)	1.00	5.43	PARK2(5.43)	
7	JNK-9L(A549)	1.00	5.38	PARK2(5.38)	
8	deoxycholic-acid(A375)	1.00	5.27	PARK2(5.27)	
9	**5-iodotubercidin(A549)**	1.00	5.26	PARK2(3.00)	PINK1(2.26)
10	**BIIB-021(YAPC)**	1.00	5.18	PINK1(2.72)	PARK2(2.46)
11	WZ-3105(MDAMB231)	1.00	5.18	PARK2(5.18)	
12	AS-703026(HT29)	1.00	5.10	PINK1(5.10)	
13	**NVP-TAE226(HT29)**	1.00	5.07	PINK1(2.72)	PARK2(2.35)
14	BMS-387032(HA1E)	1.00	4.99	PARK2(4.99)	
15	withaferin-a(HA1E)	1.00	4.89	PARK2(4.89)	
16	**fulvestrant(MCF7)**	1.00	4.83	PINK1(2.59)	PARK2(2.24)
17	CGP-60474(A549)	1.00	4.79	PARK2(4.79)	
18	**HG-14-10-04(HT29)**	1.00	4.79	PINK1(2.52)	PARK2(2.27)
19	gemifloxacin(MCF7)	1.00	4.78	PARK2(4.78)	
20	mitoxantrone(NPC)*	1.00	4.76	PARK2(4.76)	
21	phorbol-myristate-acetate(U937)	1.00	4.73	PARK2(4.73)	
22	**ganetespib(YAPC)**	1.00	4.68	PINK1(2.59)	PARK2(2.09)
23	staurosporine(NPC)	1.00	4.67	PARK2(4.67)	
24	A-443654(BT20)	1.00	4.63	PARK2(4.63)	
25	**sirolimus(HCC515)**	1.00	4.62	PARK2(2.35)	PINK1(2.27)
26	**metronidazole(MCF7)**	1.00	4.52	PINK1(2.38)	PARK2(2.14)
27	daunorubicin(CD34)	1.00	4.52	PARK2(4.52)	
28	A-443654(A375)	1.00	4.46	PARK2(4.46)	
29	**GDC-0941(BT20)**	1.00	4.45	PARK2(2.24)	PINK1(2.21)
30	AT-7519(HA1E)	1.00	4.44	PARK2(4.44)	

Supplementary Figure 2. Combined profile of drug regulation on PINK1 and PARK2 expression on LINCS extracted from SPIED3 showing the top 30 compounds The outputs are ranked based on the expression levels across the 15646 drug-like compounds in the LINCS database. The results of this query (PINK1, PARK2) was cropped to show only the 30 top ranking compounds. Only the compounds predicated to co-upregulate both PINK1 and PARK2, and the 3 top-ranking compounds were subject to further literature-based scrutiny after the second step of shortlisting. Compounds in bold were taken forward into the third step. * shows an example of repeated outputs for the same compound, mitoxantrone, with different scores indicating PARK2 upregulation, as scores of the same compound were not combined on LINCS, unlike CMap.

	compound	skew	score	gene
1	**skimmianine**	1.00	0.64	PARK2(0.64)
2	**apigenin**	1.00	0.63	PARK2(0.63)
3	**hexestrol**	1.00	0.62	PARK2(0.62)
4	**thioguanosine**	1.00	0.60	PARK2(0.60)
5	**sitosterol**	1.00	0.60	PARK2(0.60)
6	**pyridoxine**	1.00	0.58	PARK2(0.58)
7	**ethotoin**	1.00	0.58	PARK2(0.58)
8	**0175029-0000**	1.00	0.58	PARK2(0.58)
9	**oxybutynin**	1.00	0.57	PARK2(0.57)
10	**fluorometholone**	1.00	0.55	PARK2(0.55)
11	**fusaric_acid**	1.00	0.53	PARK2(0.53)
12	**ampicillin**	1.00	0.52	PARK2(0.52)
13	**ellipticine***	1.00	0.51	PARK2(0.51)
14	**roxithromycin**	1.00	0.50	PARK2(0.50)
15	**clemizole**	1.00	0.47	PARK2(0.47)
16	**aminophenazone**	1.00	0.46	PARK2(0.46)
17	**harmine**	1.00	0.46	PARK2(0.46)
18	**chlorcyclizine***	1.00	0.46	PARK2(0.46)
19	**theobromine**	1.00	0.46	PARK2(0.46)
20	**remoxipride**	1.00	0.46	PARK2(0.46)
21	5252917	1.00	0.45	PARK2(0.45)
22	Prestwick-685	1.00	0.44	PARK2(0.44)
23	eucatropine	1.00	0.43	PARK2(0.43)
24	antimycin_A	1.00	0.42	PARK2(0.42)
25	Prestwick-664	1.00	0.41	PARK2(0.41)
26	tetroquinone	1.00	0.40	PARK2(0.40)
27	GW-8510	1.00	0.38	PARK2(0.38)
28	dipyridamole	1.00	0.37	PARK2(0.37)
29	Prestwick-674	1.00	0.37	PARK2(0.37)
30	medrysone	1.00	0.37	PARK2(0.37)

Supplementary Figure 3. Profile of PARK2 regulation by compounds on CMap extracted from SPIED3 showing the top 30 compounds The outputs are ranked based on the expression levels across the 1309 drug-like compounds in the CMap database. The results of this query (PARK2 only) was cropped to show only the 30 top ranking compounds. The top 20 ranking compounds were subject to further literature-based scrutiny after the second step of shortlisting (cutoff score for PARK2 mono-upregulation = 0.46 shown by red line). Compounds in bold were taken forward into the third step. * indicates two compounds (ellipticine and chlorcyclizine) that already appeared in the combined profile (Supplementary Fig Cc and were included in the list of 33 compounds for shortlisting in second step.